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ABSTRACT

Recruitment in the information technology sector requires candidates and recruiters to process large amounts of heterogeneous information from curricula vitae and job descriptions. Conventional job portals primarily support posting, searching, and manual application workflows, while automated screening tools may provide limited transparency regarding how a score is produced or how an automated action is controlled. This thesis presents CareerFit IT AutoPilot, a web-based recruitment platform that combines a job portal, CV–job description matching, personalized job recommendation, policy-driven automation, and Human-in-the-Loop interaction in a unified workflow.

The system processes candidate profiles, curricula vitae, job descriptions, preferences, and user feedback. Textual information is normalized and represented with Term Frequency–Inverse Document Frequency vectors. Cosine similarity ranks CV–job pairs and candidate–job recommendations, while the Rocchio relevance-feedback method updates learned representations from explicit feedback. An AutoFit policy layer evaluates scores together with consent, thresholds, interaction history, quotas, cooldown rules, time zones, and quiet hours before selecting an action. Expiring email-action links use hashed token storage, a non-mutating confirmation request, and POST redemption. Audit records and action states support traceability and replay prevention.

CareerFit is implemented as a role-based web platform for guests, candidates, recruiters, and administrators using a Spring Boot backend, a React and TypeScript frontend, and PostgreSQL persistence managed through Flyway migrations. The refreshed backend suite passed 72 of 72 tests, the production frontend build completed with a largest JavaScript chunk of 378.44 kB, and all 20 Chromium workflow, contract, and resilience tests passed. On a synthetic causal benchmark containing 50 Jobs and 100 CVs, Rocchio increased nDCG@5 from 0.037737 to 0.837737. The final benchmark log contained no optimistic-lock exception, and aggregate runtime health returned HTTP 200 with status UP. These results demonstrate controlled feedback behavior and integrated prototype workflows, not production hiring effectiveness.

The project demonstrates a practical approach to recruitment automation in which explainable scoring and configurable policies assist users without removing human oversight from consequential decisions. Its current limitations include the use of lexical vector representations and controlled evaluation data; semantic representations, larger real-world studies, and extended governance mechanisms are identified as future work.
Keywords: recruitment automation; CV–job matching; recommendation system; Human-in-the-Loop; Rocchio feedback; explainable automation.

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